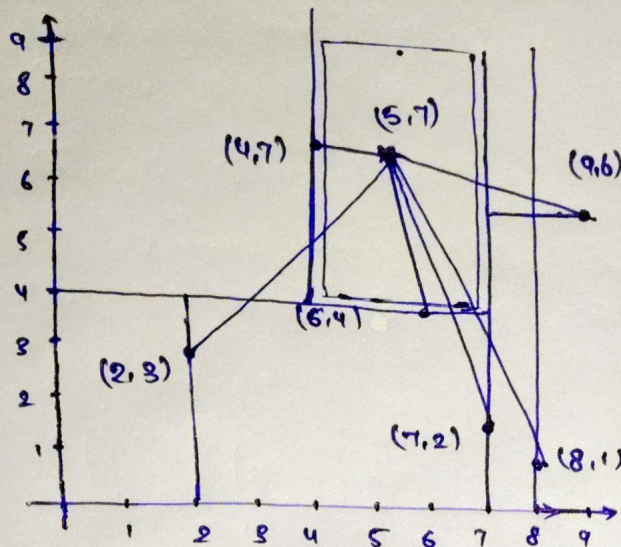


* Variants of KNN

i. KD Tree



Time Complexity
↓ ↓

d_1	d_2
7	2
5	4
9	6
2	3
4	7
8	1
Median	Median

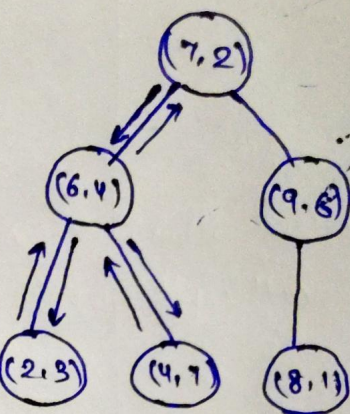
d_1
2, 4, 5, 7, 8, 9

$$\frac{5+7}{2} = \frac{12}{2} = 6.5$$

d_2
1, 2, 3, 4, 6, 7

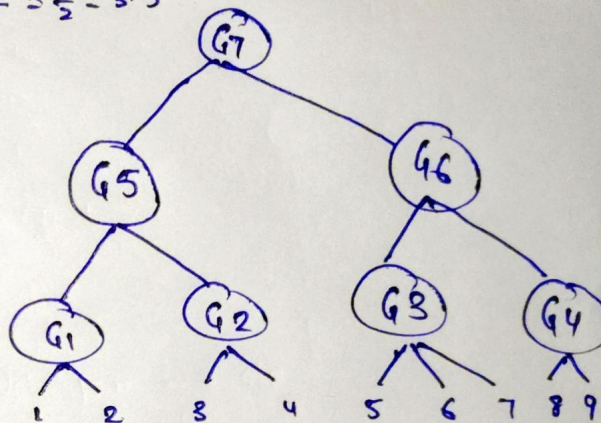
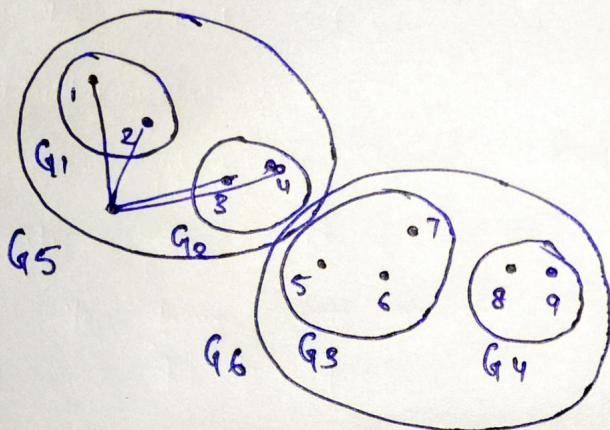
$$\frac{3+4}{2} = \frac{7}{2} = 3.5$$

Binary Tree [KD Tree]



Back Tracking

ii. Ball Tree



Note: With respect to Binary Tree the time complexity to search the element or find the distance (first search and then calculate the distance) is reduced because here we do not need to calculate the distance with each and every point.